

Handout 2

Chapter 2

ECONOMIC ACTIVITIES: PRODUCING AND TRADING

THE PRODUCTION POSSIBILITIES FRONTIER

Production Possibilities Frontier (PPF) represents the possible combinations of two goods that can be produced in a certain period of time under the conditions of a given state of technology and fully employed resources. **The Bowed-Outward (Concave-Downward) PPF:**

Increasing Opportunity Costs Assumptions:

1. Only two goods can be produced in an economy: computers and television sets.
2. As more of one good is produced, the opportunity cost between computers and television sets changes.

In Exhibit 2(a), we have identified four combinations of computers and television sets that can be produced in our economy. For example, combination A is 50,000 computers and 0 television sets, combination B is 40,000 computers and 20,000 television sets, and so on. We plotted these four combinations of computers and television sets in Exhibit 2(b). Each combination represents a different point. The curved line that connects points A–D is the production possibilities frontier.

exhibit 2

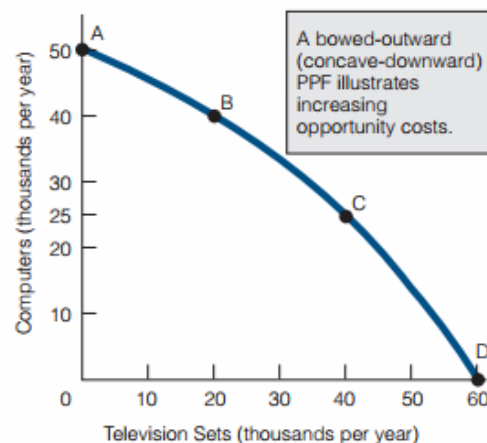
Production Possibilities Frontier (Increasing Opportunity Costs)

The economy can produce any of the four combinations of computers and televisions sets in part (a). We have plotted these combinations in part (b). The production possibilities

frontier in part (b) is bowed outward because the opportunity cost of producing television sets increases as more television sets are produced.

Combination	Computers	Television Sets	Point in Part (b)
A	50,000	0	A
B	40,000	20,000	B
C	25,000	40,000	C
D	0	60,000	D

(a)



(b)

In this case, the production possibilities frontier is bowed outward (concave downward) because the opportunity cost of television sets increases as more sets are produced

Bowed-outward PPF = Increasing opportunity costs

At point A, the economy is producing 50,000 computers and 0 television sets, and move to point B, where the economy is producing 40,000 computers and 20,000 television sets. What is the opportunity cost of a television set over this range? We see that 20,000 more television sets are produced by moving from point A to point B but at the cost of 10,000 computers. This means for every 1 television set produced, $1/2$ computer is forfeited. Thus, the opportunity cost of 1 television set is $1/2$ computer. Now let's move from point B, where the economy is producing 40,000 computers and 20,000 television sets, to point C, where the economy is producing 25,000 computers and 40,000 television sets.

Point B: 40,000 computers, 20,000 television sets; Point C: 25,000 computers, 40,000 television sets. What is the opportunity cost of a television set over this range? In this case, 20,000 more television sets are produced by moving from point B to point C but at the cost of 15,000 computers. This means for every 1 television set produced, $3/4$ computer is forfeited. Thus, the opportunity cost of 1 television set is $3/4$ of a computer.

Obviously, as the economy produces more television sets, the opportunity cost of producing television sets increases. This gives us the bowed-outward production possibilities frontier in Exhibit 2(b).

Obviously, as the economy produces more television sets, the opportunity cost of producing television sets increases. This gives us the bowed-outward production possibilities frontier in Exhibit 2(b).

The Straight-Line PPF: Constant Opportunity Costs

Assumptions:

1. Only two goods can be produced in an economy: computers and television sets.
2. The opportunity cost of one television set is one computer.
3. As more of one good is produced, the opportunity cost between television sets and computers is constant.

exhibit 1

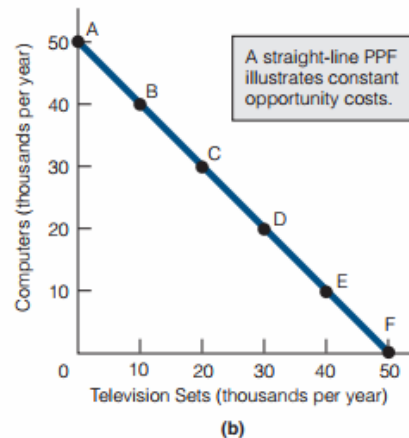
Production Possibilities Frontier (Constant Opportunity Costs)

The economy can produce any of the six combinations of computers and television sets in part (a). We have plotted these combinations in part (b). The production possibilities

frontier in part (b) is a straight line because the opportunity cost of producing either good is constant: for every 1 computer not produced, 1 television set is produced.

Combination	Computers	Television Sets	Point in Part (b)
A	50,000	0	A
B	40,000	10,000	B
C	30,000	20,000	C
D	20,000	30,000	D
E	10,000	40,000	E
F	0	50,000	F

(a)



In Exhibit 1(a), we have identified six combinations of computers and television sets that can be produced in our economy. For example, combination A is 50,000 computers and 0 television sets, combination B is 40,000 computers and 10,000 television sets, and so on. We plotted these six combinations of computers and television sets in Exhibit 1(b).

Each combination represents a different point in Exhibit 1(b). For example, the combination of 50,000 computers and 0 television sets is represented by point A. The line that connects points A–F is the production possibilities frontier.

The production possibilities frontier is a straight line in this instance because the opportunity cost of producing computers and television sets is constant.

Straight-line PPF = Constant opportunity costs

If the economy were to move from point A to point B, from B to C, and so on, the opportunity cost of each good would remain constant at 1 for 1. At point A, 50,000 computers and 0 television sets are produced. At point B, 40,000 computers and 10,000 television sets are produced.

We conclude that for every 10,000 computers not produced, 10,000 television sets are produced—a ratio of 1 to 1. The opportunity cost—1 computer for 1 television set—that exists between points A and B also exists between points B and C, C and D, D and E, and E and F. In other words, opportunity cost is constant at 1 computer for 1 television set.

Law of Increasing Opportunity Costs

The shape of the production possibilities frontier depends on whether opportunity costs (1) are constant or (2) increase as more of a good is produced. In Exhibit 1(b), the production possibilities frontier is a straight line; in Exhibit 2(b), it is bowed outward (curved). In the real

world, most production possibilities frontiers are bowed outward. This means that **for most goods, the opportunity costs increase as more of the good is produced. This is referred to as the law of increasing opportunity costs.**

But why (for most goods) do the opportunity costs increase as more of the good is produced?

The answer is because people have varying abilities. For example, some people are better suited to building houses than other people are. When a construction company first starts building houses, it employs the people who are most skilled at house building. The most skilled persons can build houses at lower opportunity costs than others can. But as the construction company builds more houses, it finds that it has already employed the most skilled builders, so it must employ those who are less skilled at house building. These (less skilled) people build houses at higher opportunity costs. Where three skilled house builders could build a house in a month, as many as seven unskilled builders may be required to build it in the same length of time. Exhibit 3 summarizes the points in this section.

exhibit 3

A Summary Statement About Increasing Opportunity Costs and a Production

Possibilities Frontier That Is Bowed Outward (Concave Downward)

Many of the points about increasing opportunity costs and a production possibilities frontier that is bowed outward are summarized here.

We start with the assumption that not all people can build houses at the same opportunity cost.

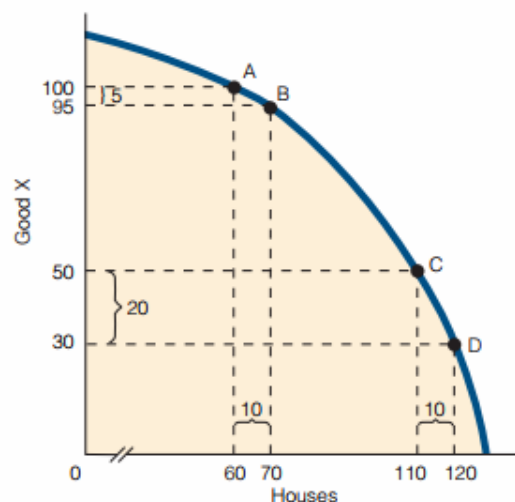
When houses are first built, only the people who can build them at (relatively) low opportunity costs will build them.

As increasingly more houses are built, people with higher opportunity costs of building houses will start building houses.

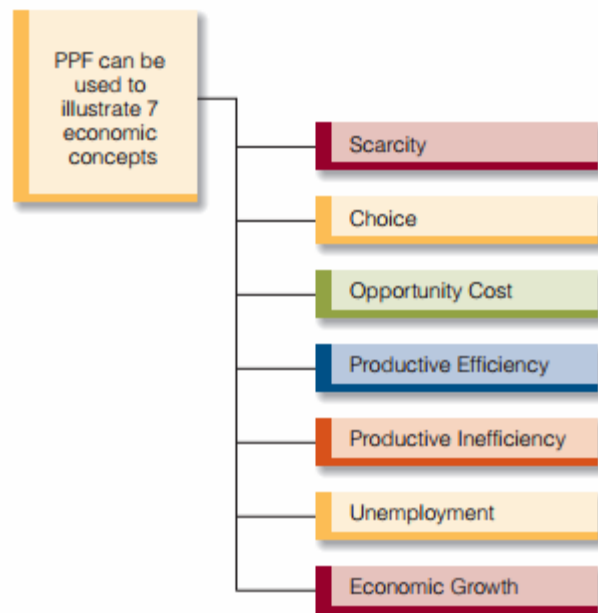
This is the same as saying that as more houses are built, the opportunity cost of building houses increases.

And this is why the PPF for houses and good X is bowed outward (concave downward). See diagram at left.

Notice that when we go from building 60 to 70 houses (10 more houses), we forfeit 5 units of good X; but when we go from building 110 to 120 houses (again, 10 more houses), we forfeit 20 units of good X.



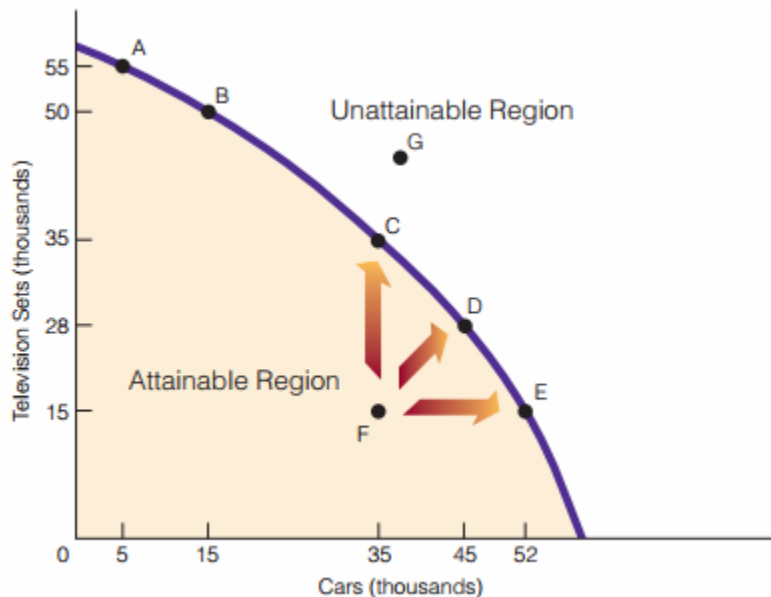
The PPF Economic Framework



Economic Concepts Within a PPF Framework

SCARCITY is the condition where wants (for goods) are greater than the resources available to satisfy those wants. The finiteness of resources is graphically portrayed by the PPF in Exhibit 5. The frontier (itself) tells us: “At this point in time, that’s as far as you can go. You cannot go any farther. You are limited to choosing any combination of the two goods on the frontier or below it.”

Exhibit 5



The PPF separates the production possibilities of an economy into two regions: (1) an attainable region, which consists of the points on the PPF itself and all points below it (this region includes points A–F) and (2) an unattainable region, which consists of the points above and beyond the

PPF (such as point G). Scarcity implies that some things are attainable and others are unattainable. Point A on the PPF is attainable, as is point F; point G is not.

Choice and opportunity cost are also shown in Exhibit 5. Note that within the attainable region, individuals must choose the combination of the two goods they want to produce. Obviously, hundreds of different combinations exist, but let's consider only two, represented by points A and B. Which of the two will individuals choose? They can't be at both points; they must make a choice.

Opportunity cost is illustrated as we move from one point to another on the PPF in Exhibit 5. Suppose we are at point A and choose to move to point B. At A, we have 55,000 television sets and 5,000 cars, and at point B, we have 50,000 television sets and 15,000 cars. What is the opportunity cost of a car? Because 10,000 more cars come at a cost of 5,000 fewer television sets, the opportunity cost of 1 car is $1/2$ television set.

Productive Efficiency is the condition where the maximum output is produced with given resources and technology. Any point on the PPF is efficient. In Exhibit 5, points A, B, C, D, and E are all productive efficient points as they lie on the PPF. In other words, we are getting the most (in terms of output) from what we have (in terms of available resources and technology).

It follows that an economy is **productive inefficient** if it is producing less than the maximum output with given resources and technology. In Exhibit 5, point F is a productive inefficient point. It lies below the production possibilities frontier; it is below the outer limit of what is possible. In other words, we could produce more goods with the resources we have available to us. Or we can get more of one good without getting less of another good.

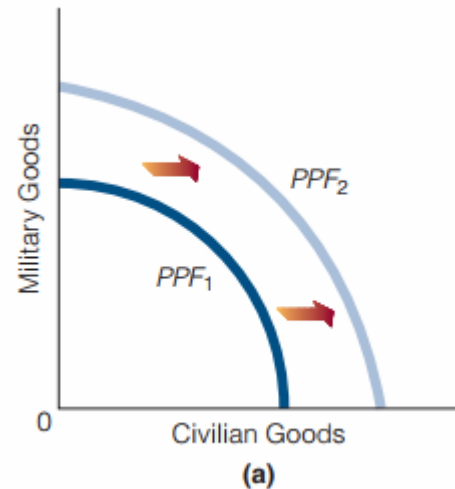
To illustrate, suppose we move from inefficient point F to efficient point C. We produce more television sets and no fewer cars. What if we move from F to D? We produce more television sets and more cars. Finally, if we move from F to E, we produce more cars and no fewer television sets. Thus, moving from F can give us more of at least one good and no less of another good. In short, productive inefficiency implies that gains are possible in one area without losses in another.

Unemployed Resources: When the economy exhibits productive inefficiency, it is not producing the maximum output with the available resources and technology. One reason may be that the economy is not using all its resources; that is, some of its resources are unemployed, as at point F in Exhibit 5. When the economy exhibits productive efficiency, it is producing the maximum output with the available resources and technology. This means it is using all its resources to produce goods; its resources are fully employed, and none are unemployed. At the productive efficient points A–E in Exhibit 5, there are no unemployed resources.

Economic growth refers to the increased productive capabilities of an economy. It is illustrated by a shift outward in the production possibilities frontier. Two major factors that affect economic growth are (1) an increase in the quantity of resources and (2) an advance in technology.

With an increase in the quantity of resources (e.g., through a new discovery of resources), it is possible to produce a greater quantity of output. In Exhibit 6, an increase in the quantity of resources makes it possible to produce both more military goods and more civilian goods. Thus, the PPF shifts outward from PPF_1 to PPF_2 .

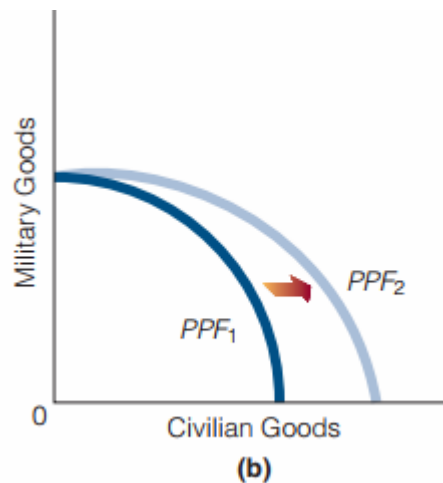
An increase in resources or an advance in technology (that can lead to more of both goods being produced) can increase the production capabilities of an economy, leading to economic



growth and a shift outward

In the PPF, as shown in part (b), If the advance in technology leads to the greater production of only one good (such as civilian goods in this exhibit 7), then the PPF shifts rotates outward, as shown in (b).

Exhibit 7



Technology refers to the body of skills and knowledge concerning the use of resources in production. An advance in technology commonly refers to the ability to produce more output with a fixed quantity of resources or the ability to produce the same output with a smaller quantity of resources.

Suppose an advance in technology allows more military goods and more civilian goods to be produced with the same quantity of resources. As a result, the PPF in Exhibit 6 shifts outward

from PPF_1 to PPF_2 . The outcome is the same as when the quantity of resources is increased. **SPECIALIZATION AND TRADE**

A country that specializes in the production of certain goods and then trades those goods to countries for other goods can make itself better off. In terms of its PPF, it can consume at a level beyond its PPF.

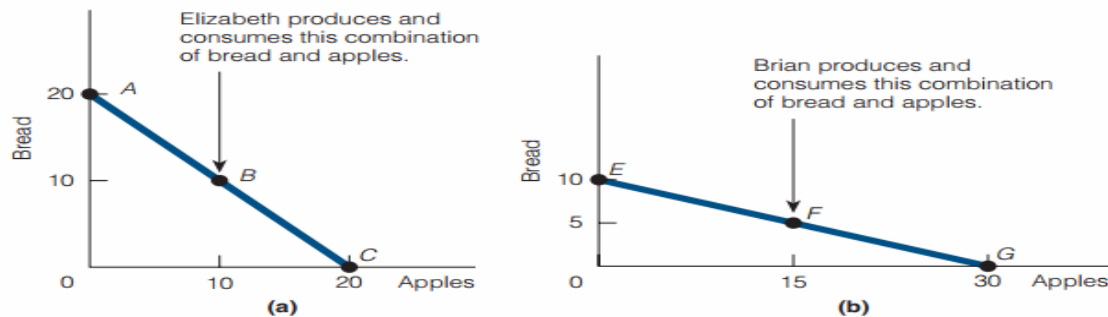
A Simple Two-Person PPF Model

Two individuals, Elizabeth and Brian engage in two activities: baking bread and growing apples. Let's suppose that, within a certain period, Elizabeth can produce 20 loaves of bread and no apples, or 10 loaves of bread and 10 apples, or no bread and 20 apples.

Elizabeth		Brian	
Bread	Apples	Bread	Apples
20	0	10	0
10	10	5	15
0	20	0	30

As a consumer, Elizabeth likes to eat both bread and apples; so she decides to produce (and consume) 10 loaves of bread and 10 apples. This combination is represented by point B in Exhibit 9(a).

Exhibit 9



During the same period, Brian can produce 10 loaves of bread and no apples, or 5 loaves of bread and 15 apples, or no bread and 30 apples. In other words, these three combinations correspond, respectively, to three points on Brian's PPF. Brian, like Elizabeth, likes to eat both bread and apples; so he decides to produce and consume 5 loaves of bread and 15 apples. This combination is represented by point F in Exhibit 9(b).

Elizabeth thinks that both she and Brian may be better off if each specializes in producing only one of the two goods and trading it for the other. In other words, Elizabeth should produce either bread or apples but not both. Brian thinks that this may be a good idea but is not sure which good each person should specialize in producing. An economist would advise each to produce the good that he or she can produce at a lower cost. In economics, a person who can produce a good at a lower cost than another person is said to have a comparative advantage in the production of the good. Exhibit 9 shows that, for every 10 units of bread Elizabeth does not

produce, she can produce 10 apples. In other words, the opportunity cost of producing 1 loaf of

$$\text{Opportunity costs for Elizabeth: } 1B = 1A$$

bread (B) is 1 apple (A):

$$1A = 1B$$

For every 5 loaves of bread that Brian does not produce, he can produce 15 apples. So, for every 1 loaf of bread he does not produce, he can produce 3 apples. Therefore, for every 1 apple he chooses to produce, he forfeits 1/3 loaf of bread:

$$\text{Opportunity costs for Brian: } 1B = 3A$$

$$1A = 1/3B$$

Comparing opportunity costs, we see that Elizabeth can produce bread at a lower opportunity cost than Brian can. (Elizabeth forfeits 1 apple when she produces 1 loaf of bread, whereas Brian forfeits 3 apples for 1 loaf of bread.) By contrast, Brian can produce apples at a lower opportunity cost than Elizabeth can. We conclude that Elizabeth has a comparative advantage in the production of bread and Brian has a comparative advantage in the production of apples. Suppose both individuals specialize in the production of the good in which they have a comparative advantage. Then Elizabeth produces only bread and makes 20 loaves. Brian produces only apples and grows 30 of them.

Now suppose that Elizabeth and Brian decide to trade 8 loaves of bread for 12 apples. In other words, Elizabeth produces 20 loaves of bread and then trades 8 of them for 12 apples. After the trade, Elizabeth consumes 12 loaves of bread and 12 apples. Compare this situation with what she consumed when she didn't specialize and didn't trade. In that situation, she consumed 10 loaves of bread and 10 apples. Clearly, Elizabeth is better off when she specializes and trades than when she does not. But what about Brian? He produces 30 apples and trades 12 of them to Elizabeth for 8 loaves of bread. In other words, he consumes 8 loaves of bread and 18 apples. Compare this situation with what he consumed when he didn't specialize and didn't trade. In that situation, he consumed 5 loaves of bread and 15 apples. Thus, Brian is also better off when he specializes and trades than when he does not.

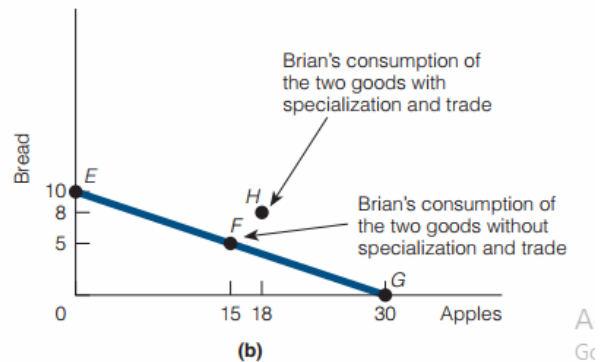
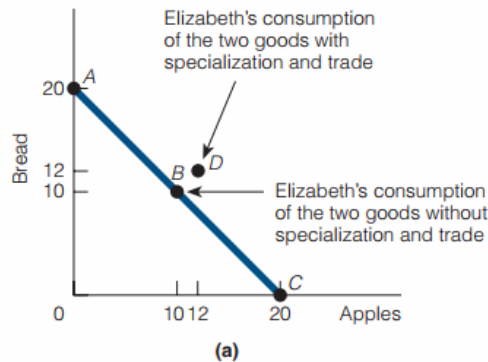
On or Beyond the PPF?

In Exhibit 10(a), we show the PPF for Elizabeth. When she was not specializing and not trading, she consumed the combination of bread and apples represented by point B (10 loaves of bread and 10 apples). When she did specialize and trade, her consumption of both goods increased, moving her to point D (12 loaves of bread and 12 apples). Here is the lesson learned: Through specialization and trade, Elizabeth's consumption moved beyond her PPF. It is easy to see the benefits of specialization and trade.

EXHIBIT 10**Consumption for Elizabeth and Brian With and Without Specialization and Trade**

A comparison of the consumption of bread and apples before and after specialization and trade shows that both Elizabeth and

Brian benefit from producing the good in which each has a comparative advantage and trading for the other good.



In Exhibit 10(b), we show the PPF for Brian. When he was not specializing and not trading, he consumed the combination of bread and apples represented by point F (5 loaves of bread and 15 apples). When he did specialize and trade, his consumption of both goods increased, moving him to point H (8 loaves of bread and 18 apples). Here is this lesson learned: Through specialization and trade Brian's consumption moved beyond his PPF.

Points to Remember

1. The PPF grounds us in reality. It tells us what is and what is not possible in terms of producing various combinations of goods and services.
2. The PPF tells us that when we have efficiency (i.e., when we are at a point on the frontier itself), more of one thing means less of something else. In other words, the PPF tells us life has its trade-offs.
3. If the PPF shifts rightward and the population do not change, then output per capita rises.

Possible questions for Quiz1 from chapter-2**1. What do you mean by a production possibility frontier?**

Production Possibilities Frontier (PPF) represents the possible combinations of two goods that can be produced in a certain period of time under the conditions of a given state of technology and fully employed resources.

2. What does bowed-outward PPF imply?

Increasing opportunity costs

3. What do you mean by the law of increasing opportunity cost?

For most goods, the opportunity costs increase as more of the good is produced. This is referred to as the law of increasing opportunity costs.

4. What does straight line PPF imply?

Constant opportunity cost.

5. What do you mean by productive efficiency?

The condition where the maximum output is produced with given resources and technology

6. What do you mean by productive inefficiency?

If an economy is producing less than the maximum output with given resources and technology.

7. What do you mean by economic growth?

Economic growth refers to the increased productive capabilities of an economy.

8. What do you mean by Technology?

Technology refers to the body of skills and knowledge concerning the use of resources in production. An advance in technology commonly refers to the ability to produce more output with a fixed quantity of resources or the ability to produce the same output with a smaller quantity of resources.

9. What are the points to be remembered from PPF?

1. The PPF grounds us in reality. It tells us what is and what is not possible in terms of producing various combinations of goods and services.
2. The PPF tells us that when we have efficiency (i.e., when we are at a point on the frontier itself), more of one thing means less of something else. In other words, the PPF tells us life has its trade-offs.
3. If the PPF shifts rightward and the population do not change, then output per capita rises.

Possible questions for Mid1 from chapter-2

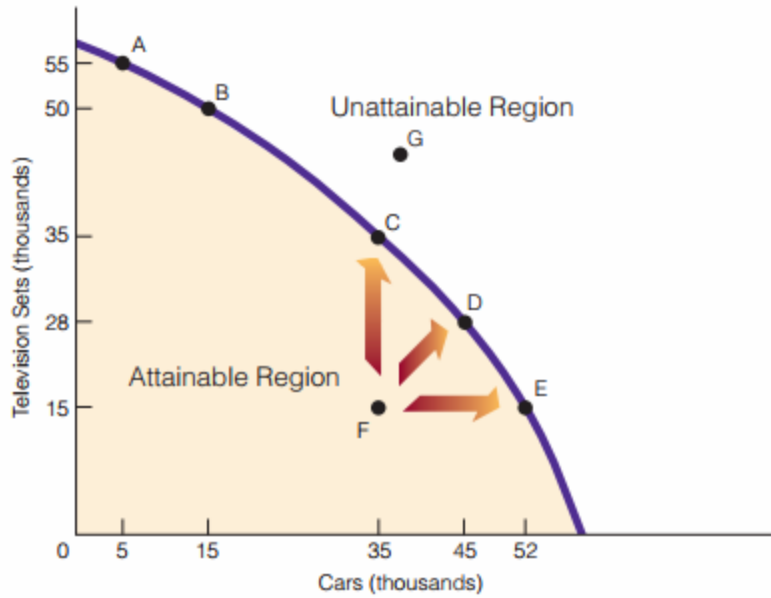
1. How do we learn 7 important economic concepts from PPF? Explain. When does PPF rotate and shift? (use graph when applicable).

We learn following 7 economic concepts from PPF:

- (a) Scarcity (b) Choice (c) Opportunity cost (d) Productive efficiency (e) Productive inefficiency (f) Unemployment (g) Economic growth

SCARCITY is the condition where wants (for goods) are greater than the resources available to

satisfy those wants. The PPF separates the production possibilities of an economy into two



regions: (1) an attainable region, which consists of the points on the PPF itself and all points below it (this region includes points A–F) and (2) an unattainable region, which consists of the points above and beyond the PPF (point G). Scarcity implies that some things are attainable and others are unattainable. Point A on the PPF is attainable, as is point F; point G is not.

Choice and opportunity cost are also shown in the figure. Note that within the attainable region, individuals must choose the combination of the two goods they want to produce. Suppose there are only two combinations, represented by points A and B. Which of the two will individuals choose? They can't be at both points; they must make a choice.

Opportunity cost is illustrated as we move from one point to another on the PPF in Exhibit 5. Suppose we are at point A and choose to move to point B. At A, we have 55,000 television sets and 5,000 cars, and at point B, we have 50,000 television sets and 15,000 cars. What is the opportunity cost of a car? Because 10,000 more cars come at a cost of 5,000 fewer television sets, the opportunity cost of 1 car is $1/2$ ($= 5000/10000$) television set.

Productive Efficiency is the condition where the maximum output is produced with given resources and technology. Any point on the PPF is efficient. Points A, B, C, D, and E are all productive efficient points as they lie on the PPF. In other words, we are getting the most (in terms of output) from what we have (in terms of available resources and technology).

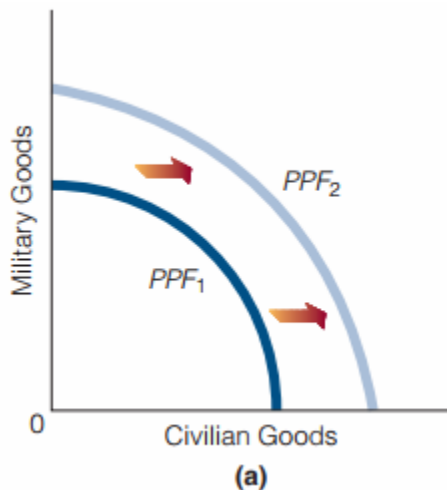
It follows that an economy is **productive inefficient** if it is producing less than the maximum output with given resources and technology. Point F is a productive inefficient point. It lies below the production possibilities frontier that implies that economy cannot use all its resources.

Unemployed Resources: When the economy exhibits productive inefficiency, it is not producing the maximum output with the available resources and technology. One reason may be that the economy is not using all its resources; that is, some of its resources are unemployed, as at point F.

Economic growth refers to the increased productive capabilities of an economy. It is illustrated by a shift outward in the production possibilities frontier. Two major factors that affect economic growth are (1) an increase in the quantity of resources and (2) an advance in technology.

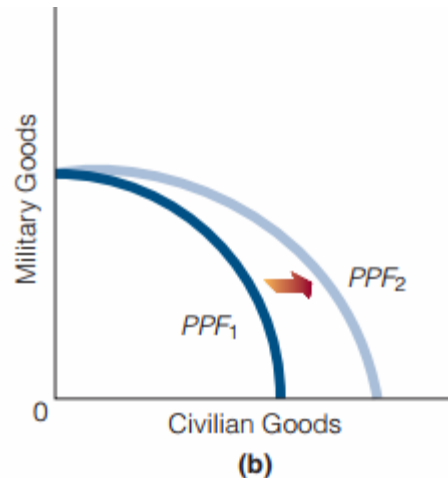
With an increase in the quantity of resources (e.g., through a new discovery of resources), it is possible to produce a greater quantity of output. In the following figure, an increase in the quantity of resources makes it possible to produce both more military goods and more civilian goods. Thus, the PPF shifts outward from PPF_1 to PPF_2 .

An increase in resources or an advance in technology can increase the production capabilities of an economy, leading to economic growth and a shift outward.



In the PPF, as shown in part (b), If the advance in technology leads to the greater production of only one good (such as civilian goods in the following figure), then the PPF rotates outward, as shown in (b).

Exhibit 7



2. How trade and specialization benefit all the parties involved in specialization and Trade? Explain.

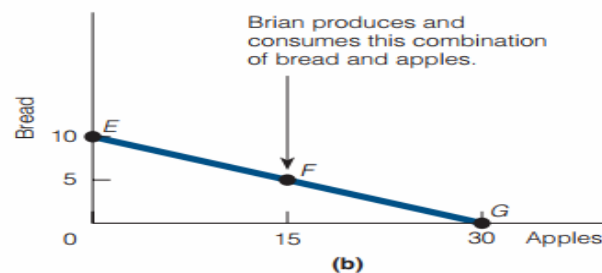
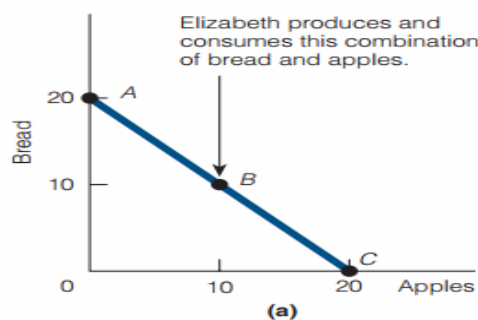
A country that specializes in the production of certain goods and then trades those goods to countries for other goods can make itself better off. In terms of its PPF, it can consume at a level beyond its PPF.

A Simple Two-Person PPF Model

Two individuals, Elizabeth and Brian engage in two activities: baking bread and growing apples. Let's suppose that, within a certain period, Elizabeth can produce 20 loaves of bread and no apples, or 10 loaves of bread and 10 apples, or no bread and 20 apples.

Elizabeth		Brian	
Bread	Apples	Bread	Apples
20	0	10	0
10	10	5	15
0	20	0	30

As a consumer, Elizabeth likes to eat both bread and apples; so she decides to produce (and consume) 10 loaves of bread and 10 apples. This combination is represented by point B in the following figure (a).



During the same period, Brian can produce 10 loaves of bread and no apples, or 5 loaves of bread and 15 apples, or no bread and 30 apples. Brian, like Elizabeth, likes to eat both bread and apples; so he decides to produce and consume 5 loaves of bread and 15 apples. This combination is represented by point F in figure(b).

Elizabeth thinks that both she and Brian may be better off if each specializes in producing only one of the two goods and trading it for the other. In other words, Elizabeth should produce either bread or apples but not both. Brian thinks that this may be a good idea but is not sure which good each person should specialize in producing. Economists suggested each of them to produce one good on the basis lower opportunity cost. Figure (a) shows that, for every 10 units of bread Elizabeth does not produce, she can produce 10 apples. In other words, the opportunity cost of producing 1 loaf of bread (B) is 1 apple (A): $10B = 10A$

$$1B = 10/10 A = 1A$$

$$1A = 1B$$

Opportunity costs for Elizabeth: $1B = 1A$

$$1A = 1B$$

For every 5 loaves of bread that Brian does not produce, he can produce 15 apples. So, for every 1 loaf of bread he does not produce, he can produce 3 apples. Therefore, for every 1 apple he chooses to produce, he forfeits $1/3$ loaf of bread:

$$5B = 15A$$

$$1B = 15/5 A = 3A$$

$$1A = 1/3 B$$

Opportunity costs for Brian: $1B = 3A$

$$1A = 1/3B$$

Comparing opportunity costs, we see that Elizabeth can produce bread at a lower opportunity cost than Brian can. (Elizabeth forfeits 1 apple when she produces 1 loaf of bread, whereas Brian forfeits 3 apples for 1 loaf of bread.) By contrast, Brian can produce apples at a lower opportunity cost than Elizabeth can. We conclude that Elizabeth has a comparative advantage in the production of bread and Brian has a comparative advantage in the production of apples. Suppose both individuals specialize in the production of the good in which they have a comparative advantage. Then Elizabeth produces only bread and makes 20 loaves. Brian produces only apples and grows 30 of them.

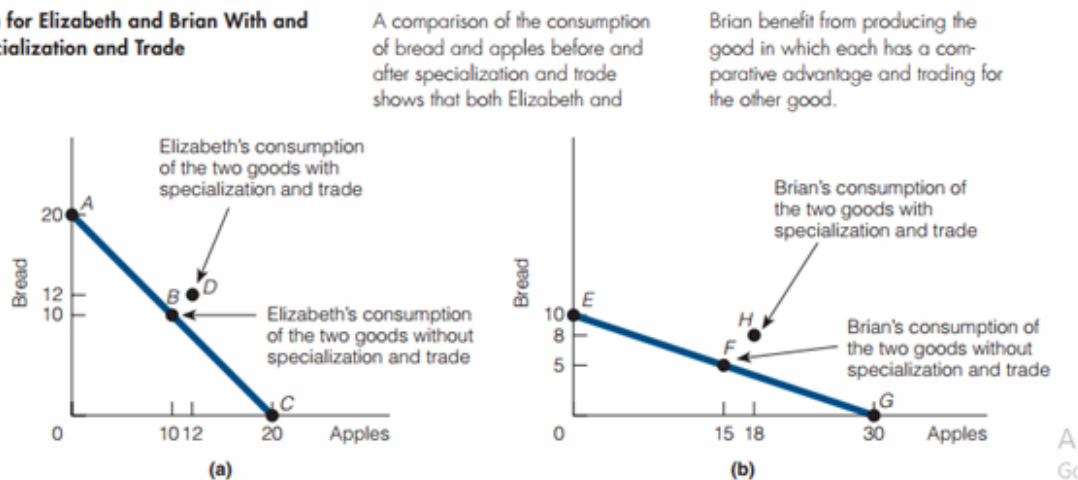
Now suppose that Elizabeth and Brian decide to trade 8 loaves of bread for 12 apples. In other words, Elizabeth produces 20 loaves of bread and then trades 8 of them for 12 apples. After the trade, Elizabeth consumes 12 loaves of bread and 12 apples. She now consumes 2 more bread

and 2 more apple than before. She gains 2B and 2A and she is now better-off. Brian produces 30 apples and trades 12 of them to Elizabeth for 8 loaves of bread. In other words, he consumes 8 loaves of bread and 18 apples. That is more 3A and 3B. He is also better-off. Both of them are better-off through specialization and trade.

On or Beyond the PPF?

In figure 10(a), we show the PPF for Elizabeth. When she was not specializing and not trading, she consumed the combination of bread and apples represented by point B (10 loaves of bread and 10 apples). When she did specialize and trade, her consumption of both goods increased, moving her to point D (12 loaves of bread and 12 apples). Here is the lesson learned: Through specialization and trade, Elizabeth's consumption moved beyond her PPF.

Consumption for Elizabeth and Brian With and Without Specialization and Trade



In figure 10(b), we show the PPF for Brian. When he was not specializing and not trading, he consumed the combination of bread and apples represented by point F (5 loaves of bread and 15 apples). When he did specialize and trade, his consumption of both goods increased, moving him to point H (8 loaves of bread and 18 apples). Here is this lesson learned: Through specialization and trade Brian's consumption moved beyond his PPF.